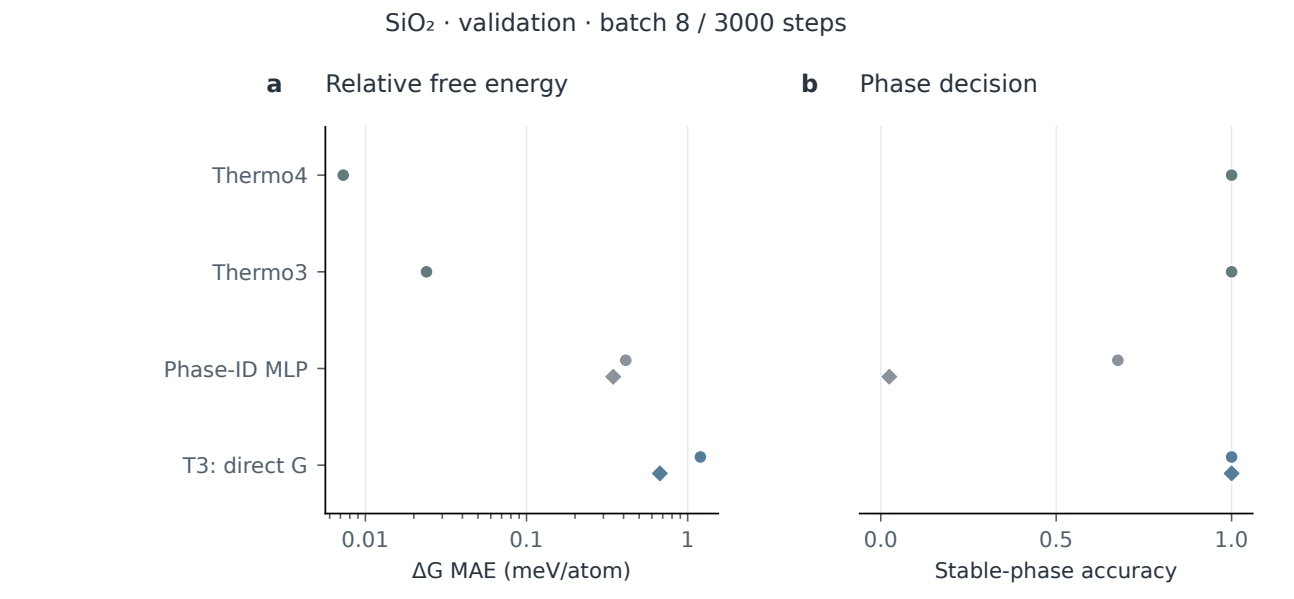
SiO₂ · validation · batch 8 / 3000 steps
a
Relative free energy
Thermo4
Thermo3
Phase-ID MLP
T3: direct G
0.01
0.1
1
ΔG MAE (meV/atom)
b
Phase decision
0.0
0.5
1.0
Stable-phase accuracy